🧠 Task Manager – Technical Documentation

1️⃣ Overview

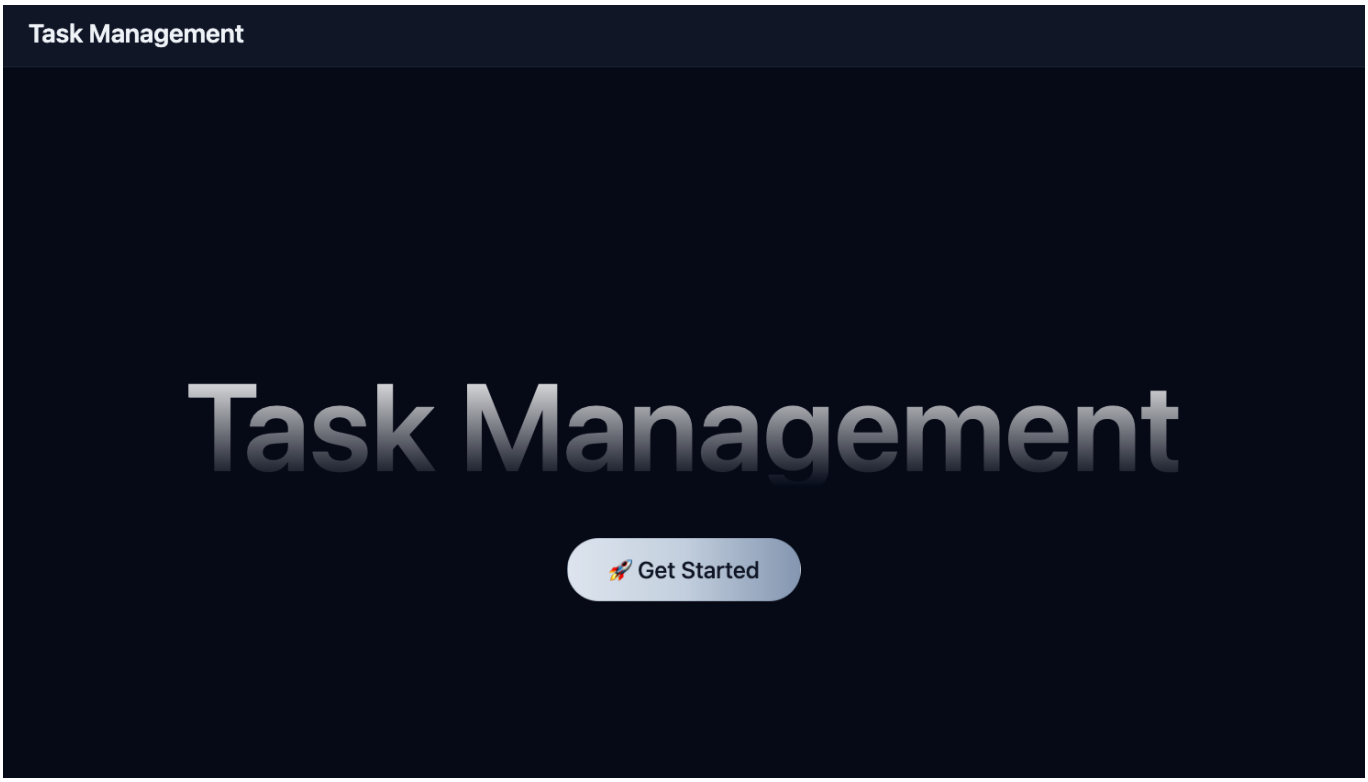
This project is a **full-stack web application** developed as part of a technical assessment.
It allows users to **create**, **list**, **delete**, and **mark tasks as completed**, with a clean, responsive, and modern interface.

The focus was on:

- Building a functional and consistent API using **Spring Boot**
- Designing a minimalist and pleasant **React** frontend using **Shadcn/UI + Tailwind**
- Ensuring **clean architecture**, **scalability**, and **code readability**

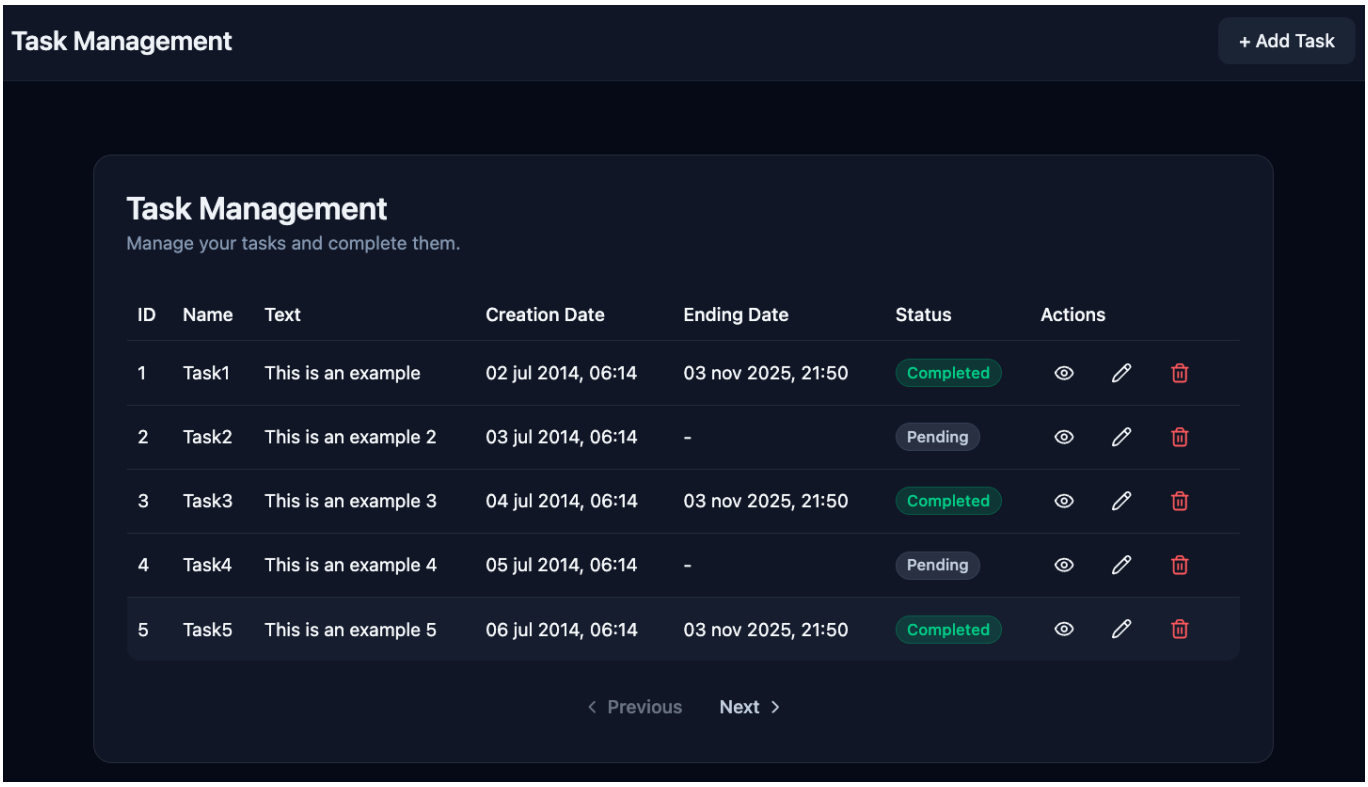
📸 Screenshots

Home Page



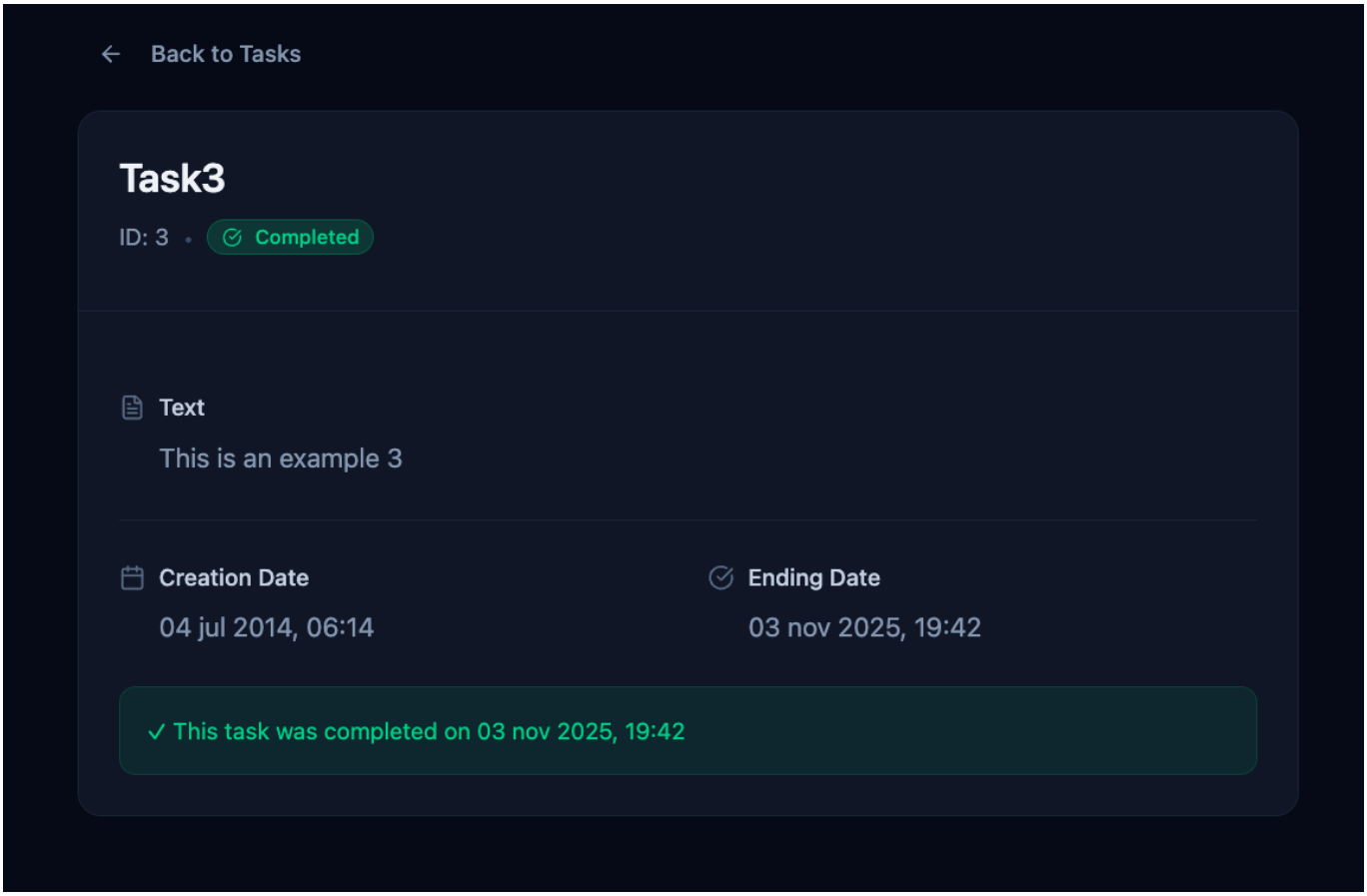
View with gradient title and "Get Started" button.

Task List



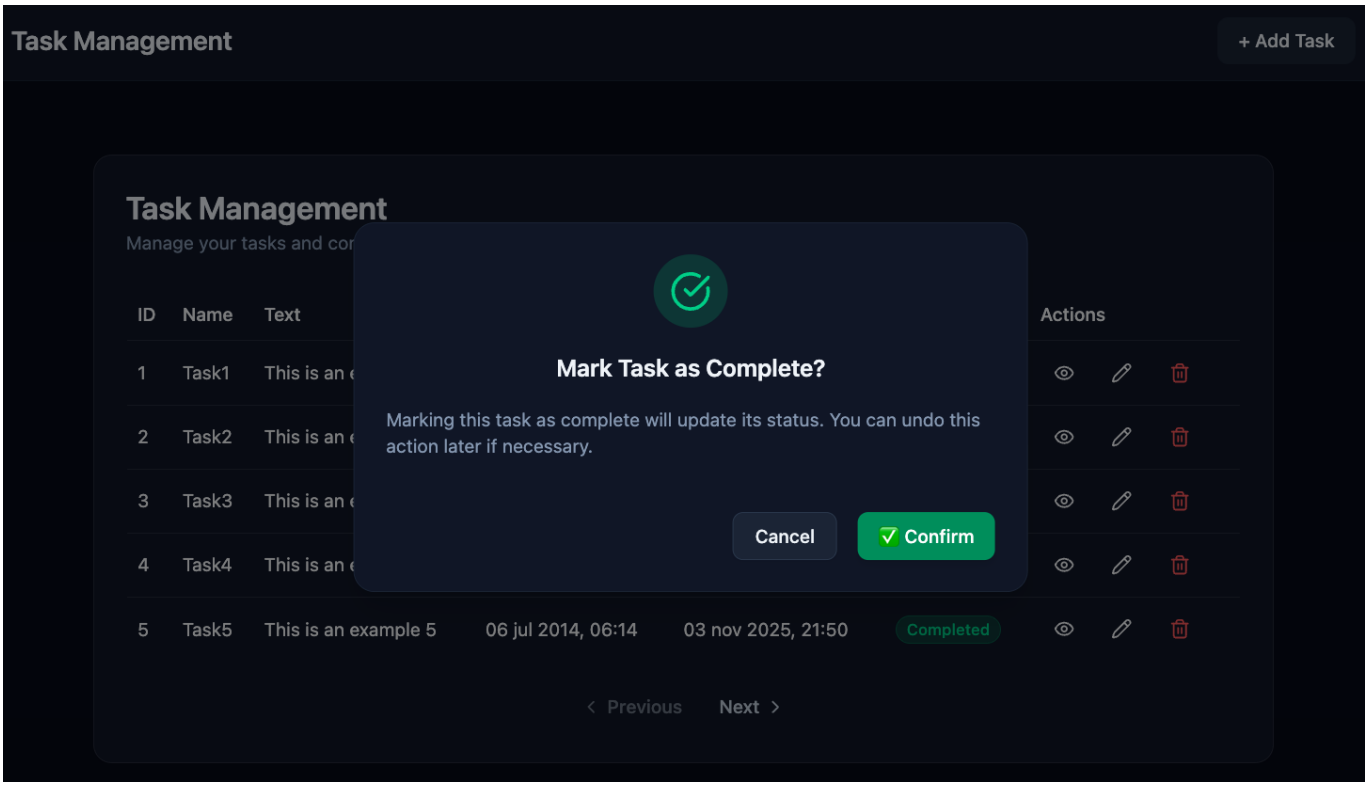
Displays all tasks with pagination and completion badges.

Task Detail



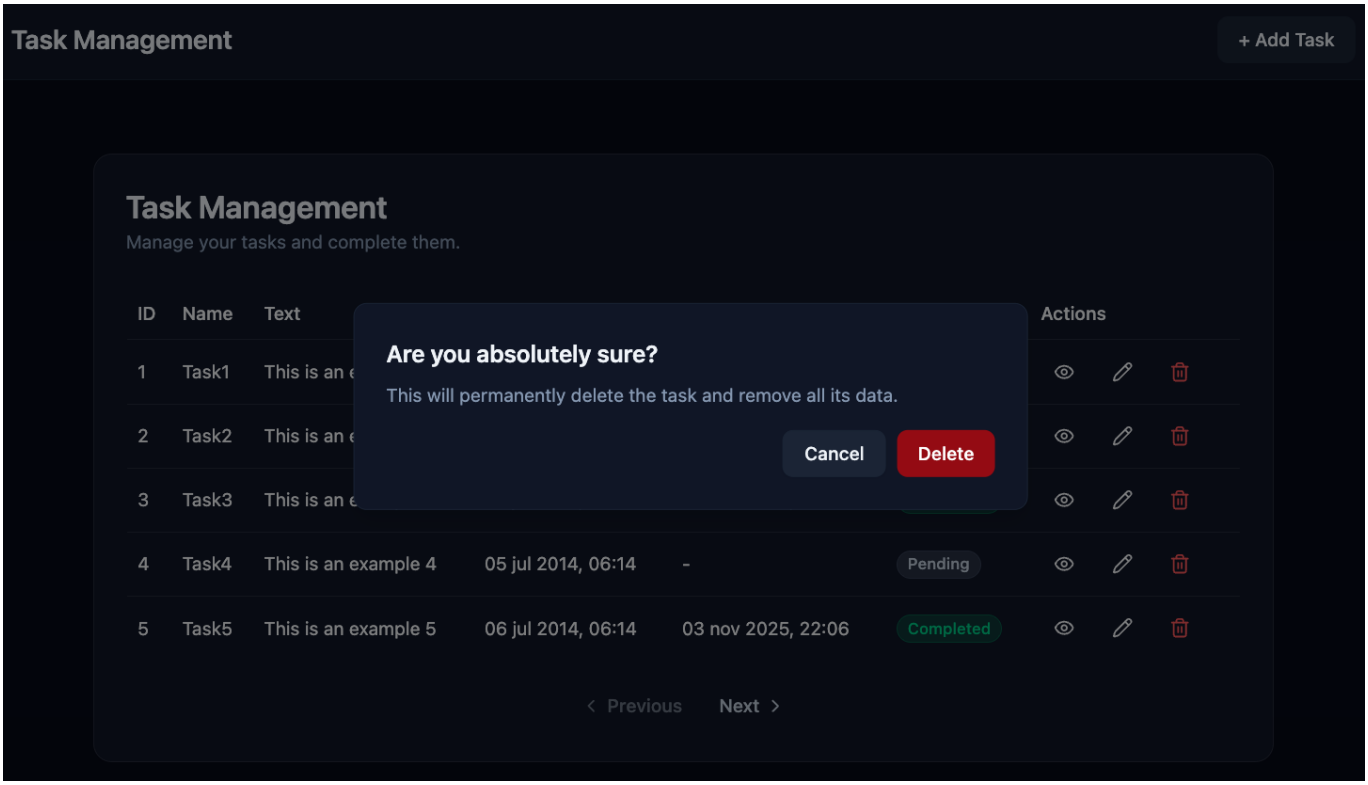
View individual task information.

Mark Task as Complete



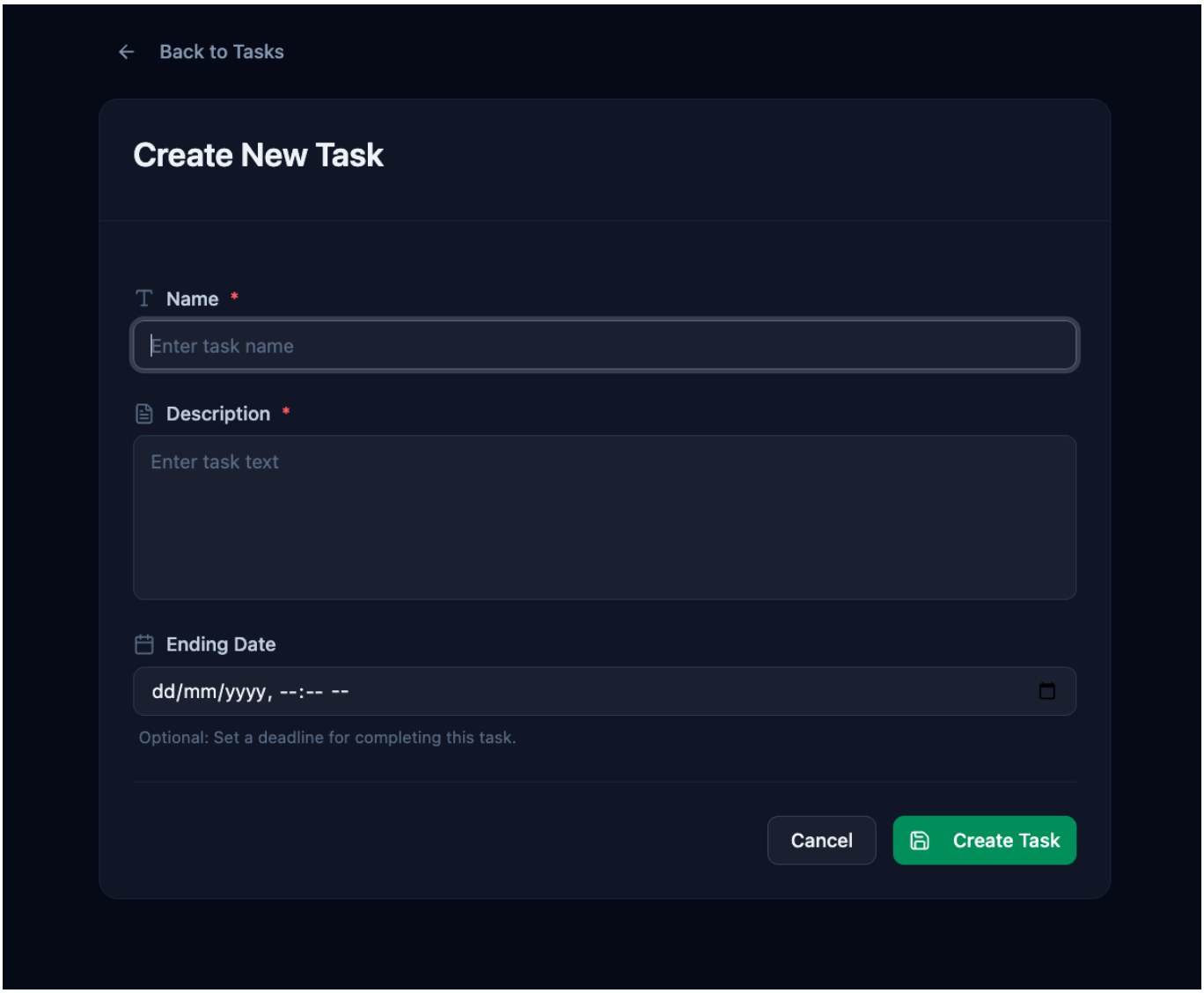
Confirmation dialog for marking a task as completed.

Delete Task



Confirmation dialog for deleting a task.

Task Create Form



Form with validation and optional ending date input.

2️⃣ Objectives

- ✅ Develop a simple task management system (CRUD simplified)
- ✅ Store and manage data in an in-memory database (H2)
- ✅ Allow toggling completion status for tasks
- ✅ Implement responsive design and form validation
- ✅ Follow clean code and best practices

3️⃣ Technologies Used

🖥️ Backend

- **Language:** Java 11
- **Framework:** Spring Boot 3.5
- **Build Tool:** Maven
- **Database:** H2 (in-memory)
- **Testing:** JUnit 5, AssertJ
- **Tools:** Yaak (API testing)

💻 Frontend

- **Language:** JavaScript (ES6)
- **Framework:** React 19 (Vite)
- **UI Library:** Shadcn/UI
- **CSS Framework:** TailwindCSS
- **Routing:** React Router DOM
- **HTTP Client:** Axios
- **Icons:** Lucide React

4️⃣ Installation & Setup

Prerequisites

- Java 11 or higher
- Node.js 18+ and npm
- Maven 3.6+

Backend Setup

```
cd backend
mvn clean install
mvn spring-boot:run
```

The API will be available at `http://localhost:8080`

Frontend Setup

```
cd frontend
npm install
npm run dev
```

The application will be available at `http://localhost:3000`

Database

H2 console available at: `http://localhost:8080/h2-console`

- JDBC URL: `jdbc:h2:mem:testdb`
- Username: `imatia`
- Password: `imatia1234`

5️⃣ Application Requirements

Requirement	Description

Requirement	Description
Create Task	The user can create a task with name, description, and optional ending date.
List Tasks	The system displays all existing tasks with pagination.
Delete Task	Tasks can be deleted by the user.
Mark as Completed	Tasks can be marked as completed, automatically assigning an ending date.
Data Validation	The backend validates name and text fields (non-null, length).
Error Handling	Proper exceptions are returned for invalid or missing data.
CORS	Frontend and backend communication enabled for localhost (React + API).

6️⃣ Architecture Overview

Backend Layers

- **Controller:** Handles HTTP requests (`TaskController.java`)
- **Service:** Business logic (`TaskService.java`)
- **DAO (Data Access Object):** Persistence layer with direct database operations (`TaskDao.java`)
- **Entity:** Data model mapped to database (`Task.java`)
- **DTO + Converter:** Transfer objects for clean layer separation

Frontend Components

Component	Description
Home.jsx	Landing page with gradient title and “Get Started” button
Navbar.jsx	Navigation bar with app branding and button to add a new task
TaskList.jsx	Displays paginated list of tasks, including completion badge
TaskCreate.jsx	Form to create a new task (with required validation)
TaskDetail.jsx	Shows detailed view of a single task with all properties
TaskService.js	Axios API requests to backend endpoints

7️⃣ API Endpoints

La API REST expone los siguientes endpoints:

Method	Route	Description
GET	`/api/tasks`	Retrieve all tasks
GET	`/api/tasks/{id}`	Retrieve a task by ID
DELETE	`/api/tasks/{id}`	Delete a task by ID

Method	Route	Description
POST	`/api/tasks`	Create a new task
PATCH	`/api/tasks/{id}/complete`	Update completion status

📦 Example JSON (POST /api/tasks)

```
{
  "name": "Finish report",
  "text": "Write technical documentation",
  "endingDate": "2025-11-04T18:00:00"
}
```

📦 Example JSON (PATCH /api/tasks/id/complete)

```
{
  "id": 1,
  "name": "Task1",
  "text": "This is an example",
  "creationDate": "2014-07-02T06:14:00.742",
  "endingDate": "2025-11-03T19:54:10.432083",
  "completed": true
}
```

📦 Example JSON (GET /api/tasks)

```
[
  {
    "id": 1,
    "name": "Task1",
    "text": "This is an example",
    "creationDate": "2014-07-02T06:14:00.742",
    "endingDate": null,
    "completed": false
  },
  {
    "id": 2,
    "name": "Task2",
    "text": "This is an example 2",
    "creationDate": "2014-07-03T06:14:00.742",
    "endingDate": null,
    "completed": false
  },
]
```

📦 **Example JSON (GET /api/tasks/3)**

```
{
  "id": 3,
  "name": "Task3",
  "text": "This is an example 3",
  "creationDate": "2014-07-04T06:14:00.742",
  "endingDate": null,
  "completed": false
}
```

8️⃣ CORS Configuration

CORS is configured via `CorsFilter` in the `SecurityConfig` class to allow access from the frontend (React/Vite):

```
    @Bean
    public CorsFilter corsFilter() {
        CorsConfiguration config = new CorsConfiguration();
        config.setAllowCredentials(false);
        config.setAllowedOrigins(List.of(
                "http://localhost:3000", "http://localhost:5173"
        ));

config.setAllowedMethods(List.of("GET","POST","PUT","PATCH","DELETE","OPTI
ONS"));
        config.setAllowedHeaders(List.of("*"));
        UrlBasedCorsConfigurationSource source = new
UrlBasedCorsConfigurationSource();
        source.registerCorsConfiguration("/**", config);
        return new CorsFilter(source);
    }
}
```

Configuration details:

- **Allowed Origins**: `localhost:3000` (Create React App) and `localhost:5173` (Vite)
- **Allowed Methods**: All CRUD operations plus OPTIONS for preflight requests
- **Allowed Headers**: All headers (`*`)
- **Credentials**: Disabled (`false`) as no authentication is implemented

This configuration enables seamless communication between the React frontend and Spring Boot backend during development.

9️⃣ Code Quality and Best Practices

The project follows clean code principles and modern development standards:

- **Single Responsibility**: Each class handles only one concern.
- **DTO usage**: Decouples entity and API response models.
- **Error Handling**: Custom exceptions (`ResourceNotFoundException`, `InvalidDateRangeException`).
- **Validation**: Required fields and logical checks (ending date after creation date).
- **Consistent naming**: LowerCamelCase for variables and methods.
- **Separation of concerns**: Controller → Service → DAO → Database.

Frontend:

- Reusable components (Buttons, Cards, Table, Dialog)
- Consistent color palette (`slate` shades)
- Responsive layout with Tailwind utility classes

🔟 Testing

🧪 Unit Tests (JUnit)

The project includes comprehensive unit tests that verify service logic, validation rules, and business operations using **Spring Boot Test** with an in-memory database.

Test configuration:

```
@SpringBootTest
@ActiveProfiles("test")
@Transactional
class TaskmanagerACApplicationTests {
    @Autowired
    private TaskService taskService;
}
```

Test coverage includes:

- ✅ **Task creation** - Validates correct task persistence

```
@Test
void createTaskTest() {
    Task newTask = new Task(1L, "Task 1", "Description example",
            Timestamp.valueOf(LocalDateTime.of(2025,2,3,10,0)),
            null, false);
    TaskDto taskDto = taskService.createTask(newTask);
    assertThat(taskDto.getName()).isEqualTo("Task 1");
}
```

- ✅ **Input validation** - Ensures null name and text are rejected

- ✅ **Date validation** - Prevents invalid date ranges (ending date before creation date)
- ✅ **Task retrieval** - Tests finding tasks by ID and listing all tasks
- ✅ **Completion toggle** - Verifies marking tasks as completed/incomplete
  - Sets `completed = true` and assigns `endingDate`
  - Sets `completed = false` and clears `endingDate`
- ✅ **Exception handling** - Confirms `ResourceNotFoundException` and `InvalidDateRangeException` are thrown appropriately

Key testing features:

- Uses `@Transactional` to rollback changes after each test
- Leverages AssertJ for fluent assertions
- Covers both happy paths and error scenarios

🔄 Integration Tests

End-to-end workflow validation testing the complete application flow:

1. **Create a new task** via `POST /api/tasks`
   - Verify task is persisted with correct data
   - Confirm `completed = false` and `endingDate = null`
2. **List all tasks** via `GET /api/tasks`
   - Verify the newly created task appears in the list
   - Confirm all task fields are correctly serialized
3. **Get task details** via `GET /api/tasks/{id}`
   - Retrieve specific task by ID
   - Validate all properties match the created task
4. **Mark task as completed** via `PATCH /api/tasks/{id}/complete`
   - Verify `completed` changes to `true`
   - Confirm `endingDate` is automatically assigned with current timestamp
5. **Mark task as pending** via `PATCH /api/tasks/{id}/complete`
   - Verify `completed` changes back to `false`
   - Confirm `endingDate` is cleared (set to `null`)
6. **Delete task** via `DELETE /api/tasks/{id}`
   - Verify correct deletion
   - Confirm the rest of task are correctly serialized

🧭 Frontend Testing

- Manual testing with Yaak and browser interactions
- Validation of form inputs (`required` fields)

- UI confirmation dialogs before updating task status
- Tested with React DevTools and browser console

Scalability and Performance

- **Stateless REST API**: Easily scalable horizontally.
- **Database abstraction**: Switching from H2 to MySQL/PostgreSQL requires minimal configuration.
- **Frontend pagination**: Implemented on the client side for intuitive data navigation
- **Optimized frontend**: Built with Vite for fast builds and hot reload.
- **Lightweight dependencies**: Ensures fast startup and minimal memory footprint.

1️⃣1️⃣ UX & UI Design

- **Framework**: Shadcn/UI with TailwindCSS styling.
- **Theme**: Dark mode with "slate" color palette.
- **Typography**: Clean and readable sans-serif.

User experience:

- Required field indicators (`*`)
- Responsive layout
- Confirmation dialogs with icons (Shadcn AlertDialog)
- Task status badges (`Completed` / `Pending`)
- Hover and focus states on all interactive elements
- Text overflow handling in tables and cards (truncation with ellipsis and proper wrapping) to keep content readable and layouts clean.

1️⃣2️⃣ Conclusion

This project demonstrates:

- A complete React + Spring Boot full-stack implementation
- Clean, modular and maintainable architecture
- Adherence to SOLID and REST principles
- Consistent UI/UX design and smooth usability
- Proper testing, scalability, and CORS setup

The result is a simple, elegant, and extendable web application aligned with industry best practices.

👨‍💻 Author

Pablo Rodríguez